MATH3900

## Tutorial 5

1. Determine whether each statement is *True* or *False*. Give a short justification, or a counterexample with verification. You may use results proved in the lecture. Here $P_C$ denotes Euclidean projection onto a nonempty, closed, convex set $C$.

   (a) For $C = \{x \in \mathbb{R}^2 : \|x\| \leq 1\}$, the vector $s = (0,1)^T$ is a feasible direction at $x = (1,0)^T$.

   (b) For every $u \in C$ and $v \in \mathbb{R}^d$,

   $$\|P_C(v) - u\| \leq \|v - u\|.$$

   (c) Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable. If $x^\star \in C$ satisfies

   $$\nabla f(x^\star)^T (v - x^\star) \geq 0 \qquad \text{for every } v \in C,$$

   then $x^\star$ must be a global minimizer of $f$ over $C$.

   (d) For every $u, v \in \mathbb{R}^d$, projection preserves their average:

   $$P_C\left(\frac{u+v}{2}\right) = \frac{P_C(u) + P_C(v)}{2}.$$

**Solution: (a) False.** A feasible direction must keep the entire short straight segment $x + ts$, $0 \le t \le \bar{t}$, inside $C$, for some $\bar{t} > 0$. Here

$$x + ts = (1, t)^T, \qquad \|x + ts\|^2 = 1 + t^2 > 1 \quad \text{for every } t > 0.$$

Thus every positive step leaves the disk, however small. The vector is tangent to the circle, but it is not a feasible direction under the lecture's straight-segment definition.

**(b) True.** Since $u \in C$, $P_C(u) = u$. The nonexpansiveness of projection therefore gives

$$\|P_C(v) - u\| = \|P_C(v) - P_C(u)\| \le \|v - u\| .$$

**(c) False.** Convexity of the feasible set alone is not enough. Take $C = [-1, 1]$, $f(x) = -x^2$, and $x^\star = 0$. The function is differentiable, with $f'(x) = -2x$, so

$$f'(0)(v - 0) = 0 \ge 0 \qquad \text{for every } v \in [-1, 1].$$

However, $1 \in C$ and

$$f(1) = -1 < 0 = f(0).$$

Thus 0 is not a global minimizer. The lecture's sufficient condition also requires the objective $f$ to be convex.

**(d) False.** Let $C = [0, 1]$, $u = 0$, and $v = 2$. The nearest points in the interval are

$$P_C(0) = 0, \qquad P_C(2) = 1, \qquad P_C(1) = 1.$$

Indeed, 0 and 1 already belong to $C$, while the right endpoint is closest to 2. Consequently,

$$P_C\left(\frac{0 + 2}{2}\right) = 1, \qquad \frac{P_C(0) + P_C(2)}{2} = \frac{1}{2}.$$

These are different. Projection does not increase distances, but it need not preserve averages.

2. Consider

$$\min_{x\in C} f(x), \qquad f(x) = \tfrac{1}{2}\big[(x_1-3)^2 + (x_2-1)^2\big],$$
$$C = \{x \in \mathbb{R}^2 : x_1 \geq 0,\ x_2 \geq 0,\ x_1 + x_2 \leq 2\}.$$

Let $x^\star = (2,0)^T$.

(a) Write the constraints as $g_i(x) \leq 0$, $i = 1,2,3$, in the order displayed. Identify the active inequality set $\mathcal{A}(x^\star)$.

(b) Find all feasible directions $s = (s_1, s_2)^T$ at $x^\star$. Justify both necessity and sufficiency using $x^\star + ts$.

(c) Use the first-order optimality condition over a convex set to prove that $x^\star$ is the unique global minimizer. Is $\nabla f(x^\star) = 0$? Explain what happens to the ordinary gradient step $x^\star - \alpha\nabla f(x^\star)$ for any $\alpha > 0$.

**Solution:** **(a)** In the required form,

$$g_1(x) = -x_1, \qquad g_2(x) = -x_2, \qquad g_3(x) = x_1 + x_2 - 2.$$

At $x^\star = (2,0)^T$, these values are $-2, 0, 0$, respectively. Thus $x^\star$ is feasible and

$$\mathcal{A}(x^\star) = \{2,3\}.$$

The first inequality is inactive because $g_1(x^\star) < 0$.

**(b)** Starting from $x^\star = (2,0)^T$, a step in the direction $s = (s_1, s_2)^T$ gives

$$x^\star + ts = (2 + ts_1, ts_2)^T.$$

Since the starting point lies on the boundaries $x_2 = 0$ and $x_1 + x_2 = 2$, a feasible direction cannot move outside either boundary. For $t > 0$, these two constraints require

$$ts_2 \geq 0, \qquad 2 + t(s_1 + s_2) \leq 2,$$

or equivalently,

$$s_2 \geq 0, \qquad s_1 + s_2 \leq 0.$$

It remains to check $x_1 \geq 0$, that is, $2 + ts_1 \geq 0$. The first coordinate starts at $2 > 0$, so moving to the left is allowed as long as the step is sufficiently short. Indeed, the conditions above imply $s_1 \leq -s_2 \leq 0$. If $s_1 < 0$, choosing

$$0 \leq t \leq \bar{t} := \frac{2}{-s_1}$$

ensures $2 + ts_1 \geq 0$. If $s_1 = 0$, then $0 \leq s_2 \leq -s_1 = 0$, so $s = 0$ and the point does not move.

Thus every direction satisfying the two conditions admits a positive step-size bound within which all constraints hold. The set of feasible directions is therefore

$$\boxed{\{s \in \mathbb{R}^2 : s_2 \geq 0,\ s_1 + s_2 \leq 0\}.}$$

**Solution:** **(c)** The gradient and Hessian are

$$\nabla f(x) = (x_1 - 3, x_2 - 1)^T, \qquad \nabla^2 f(x) = I.$$

Thus $f$ is strictly convex. The feasible set is convex, being an intersection of three halfspaces. At the proposed minimizer,

$$\nabla f(x^\star) = (-1, -1)^T.$$

For any $v = (v_1, v_2)^T \in C$,

$$\begin{aligned} \nabla f(x^\star)^T (v - x^\star) &= (-1, -1)\big((v_1, v_2)^T - (2, 0)^T\big) \\ &= 2 - v_1 - v_2 \geq 0. \end{aligned}$$

The last inequality uses $v_1 + v_2 \leq 2$. The convex first-order condition therefore proves that $x^\star$ is a global minimizer. Strict convexity makes it unique.

The gradient is not zero. An ordinary gradient step gives

$$x^\star - \alpha \nabla f(x^\star) = (2 + \alpha, \alpha)^T.$$

Its coordinates sum to $2 + 2\alpha > 2$, so it is infeasible for every $\alpha > 0$. Thus a nonzero gradient does not contradict constrained optimality: the ordinary negative-gradient step is not feasible here.

3. Let

$$C = \{x \in \mathbb{R}^2 : x_1 + 2x_2 \leq 2\}, \qquad u = (2,2)^T, \qquad a = (1,2)^T.$$

Use the projection characterization from the lecture: for $z \in C$,

$$z = P_C(u) \quad \Longleftrightarrow \quad (u-z)^T(v-z) \leq 0 \quad \text{for every } v \in C.$$

(a) The point $\widehat{z} = (0,1)^T$ is feasible. Use $v = (1,1/2)^T$ in the characterization to show that $\widehat{z} \neq P_C(u)$.

(b) Look for a candidate of the form $z = u - ta$ on the boundary $z_1 + 2z_2 = 2$. Find $t$ and $z$, then verify that $z = P_C(u)$ using the characterization for an arbitrary $v \in C$. Compute $\|u - z\|^2$.

(c) For the projection $z$ found in (b), expand $\|u - v\|^2$ using $u - v = (u - z) - (v - z)$. Show that

$$\|u - v\|^2 \geq \frac{16}{5} + \|v - z\|^2 \qquad \text{for every } v \in C.$$

Explain why no other feasible point is as close to $u$ as $z$.

**Solution:** **(a)** Both points are feasible, because

$$\widehat{z}_1 + 2\widehat{z}_2 = 0 + 2 = 2, \qquad v_1 + 2v_2 = 1 + 2(1/2) = 2.$$

The two displacement vectors are

$$u - \widehat{z} = (2, 1)^T, \qquad v - \widehat{z} = (1, -1/2)^T.$$

Therefore

$$(u - \widehat{z})^T (v - \widehat{z}) = 2 - \frac{1}{2} = \frac{3}{2} > 0.$$

This violates the characterization for one feasible $v$, so $\widehat{z}$ is not the projection. Feasibility alone is insufficient.

**(b)** The proposed form gives $z = (2 - t, 2 - 2t)^T$. Imposing the boundary equation yields

$$(2 - t) + 2(2 - 2t) = 2 \quad \Longrightarrow \quad 6 - 5t = 2 \quad \Longrightarrow \quad t = \frac{4}{5}.$$

Thus

$$z = (6/5, 2/5)^T, \qquad u - z = (4/5, 8/5)^T = \frac{4}{5}a.$$

This point belongs to $C$, since $6/5 + 2(2/5) = 2$. For every $v \in C$,

$$\begin{aligned}(u - z)^T (v - z) &= \frac{4}{5}\left(v_1 - \frac{6}{5} + 2\left(v_2 - \frac{2}{5}\right)\right) \\ &= \frac{4}{5}(v_1 + 2v_2 - 2) \leq 0.\end{aligned}$$

Hence the characterization verifies the candidate:

$$\boxed{P_C(u) = (6/5, 2/5)^T.} \qquad \|u - z\|^2 = \frac{16}{25} + \frac{64}{25} = \frac{16}{5}.$$

The form $z = u - ta$ suggested a candidate; the inequality proves it is the projection.

**(c)** For any $v \in C$, expanding the squared norm gives

$$\begin{aligned}\|u - v\|^2 &= \|u - z\|^2 + \|v - z\|^2 - 2(u - z)^T (v - z) \\ &\geq \|u - z\|^2 + \|v - z\|^2 = \frac{16}{5} + \|v - z\|^2.\end{aligned}$$

The term $-2(u - z)^T (v - z)$ is nonnegative because $(u - z)^T (v - z) \leq 0$. If $v \neq z$, then $\|v - z\|^2 > 0$, so $\|u - v\|^2 > 16/5 = \|u - z\|^2$. Thus $z$ is strictly closer to $u$ than every other feasible point.

4. Use the same halfspace $C$, point $u$, and projection $z = P_C(u)$ as in Question 3. Consider two perturbations of the input:

$$d_T = (1/5, -1/10)^T, \qquad d_N = (1/10, 1/5)^T, \qquad u_T = u + d_T, \quad u_N = u + d_N.$$

(a) Calculate $\|d_T\|$ and $\|d_N\|$. Without computing the new projections, use nonexpansiveness to bound $\|P_C(u_T) - z\|$ and $\|P_C(u_N) - z\|$.

(b) Verify, using the projection characterization, that

$$P_C(u_T) = z + d_T = (7/5, 3/10)^T, \qquad P_C(u_N) = z = (6/5, 2/5)^T.$$

(c) Compute the actual changes in the two projections. For which perturbation is the bound in (a) attained? Explain the difference geometrically using the boundary $x_1 + 2x_2 = 2$ and its normal vector $a = (1, 2)^T$.

**Solution:** **(a)** Both perturbations have the same size:

$$\|d_T\|^2 = \frac{1}{25} + \frac{1}{100} = \frac{1}{20}, \qquad \|d_N\|^2 = \frac{1}{100} + \frac{1}{25} = \frac{1}{20}.$$

Hence $\|d_T\| = \|d_N\| = \sqrt{5}/10$. Since $P_C(u) = z$, nonexpansiveness gives

$$\|P_C(u_T) - z\| = \|P_C(u_T) - P_C(u)\| \le \|u_T - u\| = \frac{\sqrt{5}}{10},$$
$$\|P_C(u_N) - z\| = \|P_C(u_N) - P_C(u)\| \le \|u_N - u\| = \frac{\sqrt{5}}{10}.$$

**(b)** First set $w_T = z + d_T = (7/5, 3/10)^T$. It lies on the boundary, since

$$a^T w_T = \frac{7}{5} + 2\frac{3}{10} = 2.$$

The perturbation cancels in the displacement:

$$u_T - w_T = (u + d_T) - (z + d_T) = u - z = \frac{4}{5}a.$$

For every $v \in C$,

$$(u_T - w_T)^T (v - w_T) = \frac{4}{5}(a^T v - 2) \le 0.$$

The characterization proves $P_C(u_T) = w_T$.

For the second input, use the feasible candidate $w_N = z$. Since $d_N = a/10$,

$$u_N - z = (u - z) + d_N = \frac{4}{5}a + \frac{1}{10}a = \frac{9}{10}a.$$

For every $v \in C$,

$$(u_N - z)^T (v - z) = \frac{9}{10}(a^T v - 2) \le 0.$$

The characterization gives $P_C(u_N) = z$.

**(c)** The actual changes are

$$\|P_C(u_T) - z\| = \|d_T\| = \frac{\sqrt{5}}{10}, \qquad \|P_C(u_N) - z\| = 0.$$

The first bound is attained; the second is strict. Since $a^T d_T = 1/5 - 2/10 = 0$, $d_T$ is parallel to the boundary. The input and its projection shift by the same amount along the boundary. In contrast, $d_N = a/10$ points in the outward normal direction. It moves the input farther from the halfspace without changing its nearest point. Nonexpansiveness allows both behaviours: the output distance is at most the input distance, not necessarily strictly smaller.